PA CONFIDENCE SCORE CALCULATION FOR TRIAGE

What a PA coordinator's day actually looks like (Current User Journey)

The flowchart below maps every step in the current prior authorization process. The right column annotates where time is lost, errors enter, or patients fall out. This is the baseline against which the new journey (Section 7) should be evaluated.

STEP 1 Prescriber writes Rx	*No friction at this stage.*
↓	
STEP 2 Benefits verification	*Staff checks coverage. PA required flag raised. No intelligence on which PAs will be hard.*
↓	
STEP 3 Form selection ⚠ FRICTION 1	*~15–20% of PAs use the wrong form → instant administrative denial → 3–5 days lost.*
↓	
STEP 4 Fill clinical answers ⚠ FRICTION 2	*Staff manually searches EHR. Missing data = #1 denial cause (30–35% of denials). No visibility into what is missing before submitting.*
↓	
STEP 5 Human review of AI draft ⚠ THE CORE GAP	*Every PA reviewed at the same effort level,4.5 min,regardless of AI confidence. A 94% confidence PA gets the same time as a 22% confidence PA.*
↓	
STEP 6 Submit to payer	*Develop Health automates this step well. No friction.*
↓	
STEP 7 Await determination ⚠ FRICTION 3	*7–14 day wait. No intelligence on which PAs are at risk in flight. 100% reactive.*
↓	
STEP 8 Payer decision	*Approved → patient fills Rx. Denied → 4–7 hrs appeal. 25% of patients abandon during wait.*
↓ if Denied	
STEP 9 Denial management	*Appeal or resubmit. Root cause not systematically captured. Same errors recur next month.*

> *Three friction points, one root cause: the system has no way to predict which PAs are at risk before they are submitted. Every intervention is reactive. The PA Confidence Score makes it proactive.*

The flowchart below maps the same process after the confidence score is inserted. The number of steps is identical,the difference is what happens at each friction point. Compare directly to Section 3.

STEP 1 Prescriber writes Rx	*No change.*
↓	
STEP 2 Benefits verification	*No change,PA required flag raised.*
↓	
STEP 3 AI auto-selects form + fills clinical answers	*Same as before,but now this step outputs a confidence score before the PA enters the review queue.*
↓	
STEP 4 CONFIDENCE SCORE COMPUTED ← NEW	*0–100 score assigned. Green / Yellow / Red tier determined. Explainability card generated for Yellow and Red cases.*
↓	
STEP 5a GREEN,Bulk approve	*One button. All Green PAs approved in under 30 seconds total. 53% of the queue handled before first coffee.*
STEP 5b YELLOW,Targeted review	*Explainability card shows the specific low factor and what to fix. Review time: 2–3 minutes instead of 4.5.*
STEP 5c RED,Intervention before submission	*PA is not submitted. Exact missing elements listed. Coordinator collects data, fixes, requeues. Denial prevented before it occurs.*
↓	
STEP 6 Submit to payer	*Only reviewed, complete PAs submitted. Higher first-pass approval rate.*
↓	
STEP 7 IN-FLIGHT WATCH LIST ← NEW	*Submitted Red-tier PAs stay visible. Proactive alert at 5 days if no determination. Coordinator intervenes before denial is issued.*
↓	
STEP 8 Payer determination	*Higher approval rate. Fewer denials. Fewer appeals.*
↓ FEEDBACK LOOP ← NEW	
STEP 9 Outcomes feed back into model	*Every approval and denial updates the historical approval rate. Every staff edit improves AI quality scoring. The model gets more accurate with every PA processed. No competitor can replicate this without years of volume.*

> *The most important change is not the triage queue,it is the feedback loop. Every PA processed teaches the system something. The model compounds. The moat deepens. This is the difference between a product feature and a strategic asset.*

1. Prototype Features

The PA Confidence Score & Triage UI consists of **five** interconnected features.

Feature 1: Confidence Score Computation Engine (Backend)

User story

As a product system, when a PA is submitted by the AI, I **need to compute a confidence score before it reaches the human review queue,** so that the triage layer has the signal it needs to route it correctly.

What it computes

The confidence score is a **composite 0–100 integer score derived from four input signals.** Each factor contributes a weighted sub-score. The composite is computed at PA generation time and updated if the PA is re-queued after staff edits.

The four factors and their weights

Factor	What it measures	Weight	Data source
Historical approval rate	For this exact drug × payer × plan combination, what % of previously submitted PAs were approved on first pass?	35%	Develop Health PA outcome DB
Documentation completeness	How many of the payer's known required data elements were present and parseable in the EHR at PA generation time?	30%	EHR integration + payer criteria library
AI answer quality	LLM self-confidence score + semantic coherence of the generated clinical justification + step therapy logic check	25%	LLM eval layer
Payer criteria freshness	How recently was the payer's step therapy criteria last verified? Has a payer criteria change been detected since the training data was collected?	10%	Payer criteria monitoring system

Composite formula

Confidence Score = round((0.35 × Historical_Approval_Rate) + (0.30 × Documentation_Completeness) + (0.25 × AI_Answer_Quality) + (0.10 × Payer_Criteria_Freshness))

> *The weights above are starting points, not permanent values. They should be validated against outcomes data in the first 60 days and recalibrated quarterly. The most important weight to get right is Documentation Completeness , it is the most actionable factor and the one where staff intervention has the highest leverage.*

Triage thresholds

Tier	Score range	Staff action	Target review time
Green	75–100	One-click approve or bulk approve. AI answer pre-checked as high quality.	<30 seconds
Yellow	40–74	Review AI answer. Specific attention flag surfaced for lowest sub-score.	2–3 minutes

Red	0–39	Do not submit. Fix flagged issues first. Specific missing data listed.	5–10 min (intervention)

Edge cases and rules

- If historical approval rate is **unavailable** (new drug, new payer, <10 historical submissions): cap that factor contribution at 50 and flag **'Insufficient history'** in the explainability card. Do not zero out the score, **insufficient history is not the same as low confidence.**
- If a **payer criteria change** has been detected since the last training refresh: force Payer Criteria Freshness sub-score to 0 and elevate triage tier by one level (Green → Yellow, Yellow → Red). Surface explicit 'Payer criteria changed X days ago, verify answers' flag.
- If documentation completeness is **below 40%** AND historical approval rate is below 50%: force override to Red regardless of composite score. The combination of poor docs and poor historical rate is the highest-risk scenario.
- **Score recomputation:** if a staff member edits an AI answer and requeues, **recompute only the AI Answer Quality sub-score**. Historical approval rate and documentation completeness are fixed at PA generation time.

Feature 2: Triage Queue, The Primary Staff Interface

User story

As a PA coordinator starting my morning review session, I need to **see today's PA queue sorted by confidence tier,** with the most actionable PAs at the top and a clear recommended action for each, so I can process the queue efficiently without reading every PA in full.

Layout and hierarchy

The Triage Queue is the default view when a user logs into the Develop Health platform on any given day. It replaces the current flat list view. The layout has three levels:

- **Summary bar at top:** count of Green / Yellow / Red PAs today, estimated total review time, and the Bulk Approve trigger for Green tier.
- **Filter tabs:** All / Green / Yellow / Red. Persists across sessions (if a user prefers to see only Red PAs first, the app remembers this preference).
- **PA row list**: each row is a collapsed card showing Patient identifier, Drug name, Payer, Confidence score badge (colored), and recommended action chip. Clicking a row expands the detail view.

PA row expanded state (detail view)

When a staff member clicks a row, the expanded view shows:

- **Confidence Score breakdown:** a mini bar-chart showing each of the four sub-scores with their labels. The lowest sub-score is highlighted in amber or red.
- **Explainability card (for Yellow/Red only):** 2–3 sentence explanation of why the score is low and what specific action could improve it. Example: 'Low score driven by documentation completeness (42%). Humana requires a documented A1C value within 6 months for semaglutide approval. No A1C found in EHR for this patient.'
- **AI answer preview:** the full AI-generated PA answer, scrollable inline. For Green PAs, this is collapsed by default (one-click approve is the primary action). For Yellow/Red, this is expanded by default.
- **Action buttons:** 'Approve and submit' (Green/Yellow after review), 'Edit answer', 'Add missing documentation' (deep-links to EHR field), 'Escalate to prescriber'.

Bulk Approve mechanics

The Bulk Approve button in the summary bar approves all Green-tier PAs in a single action. Before executing, a confirmation screen shows the count of PAs being approved and a 5-second undo window. After bulk approval, each PA is marked as **'Staff-reviewed' in the audit log with timestamp**. This maintains HIPAA audit trail requirements while enabling speed.

Accessibility and edge cases

- If all PAs in the queue are Red-tier on a given day (unusual, but can happen during a payer criteria change event): surface an alert: 'Unusual: 100% of today's queue is Red-tier. This may indicate a payer criteria update. Review payer alerts before processing.'
- If a staff member bulk-approves Green PAs and one of those PAs is subsequently **denied: log the failure,** exclude that drug/payer/plan/outcome from the Green tier's historical rate, and **trigger a model recalibration alert.**
- **Mobile:** the queue must be fully functional on tablet. Many PA coordinators work from tablets when moving between exam rooms.

Feature 3: Explainability Card, AI Reasoning in Plain English

User story

As a PA coordinator reviewing a Yellow-tier PA, I need to understand specifically WHY the AI flagged it, in plain English terms I understand (not data science language), so I can **decide whether to edit the answer, collect missing information, or override and submit anyway.**

Design specification

The explainability card is a 3–4 sentence structured summary, automatically generated by the system based on which sub-scores are lowest. It is NOT generated by an LLM (to avoid the irony of using AI to explain AI failures unreliably). It is generated by a deterministic template system that maps sub-score combinations to pre-written, clinically reviewed explanation templates.

Template structure

Each template follows the pattern: '[Score driver] , [What it means clinically] , [Specific data that is missing or flagged] , [Recommended action].' Examples:

> *Documentation completeness: 38%. Cigna requires documented gabapentin trial failure (300mg minimum, 4+ weeks) for pregabalin Step 1 approval. No gabapentin history found in patient EHR. Recommended: request prescriber to document gabapentin history in the clinical note, then requeue this PA.*

> *Payer criteria freshness: 0%. Humana updated their semaglutide step therapy criteria 8 days ago (new BMI threshold: 27 with comorbidity, down from 30). AI answer was generated before this update. Review the new criteria and verify the patient's BMI documentation meets the updated threshold before submitting.*

Template library scope

The initial template library should cover the top 20 denial reasons by volume across Develop Health's PA data. The PM should work with the clinical operations team to review and approve each template before launch. Templates must be reviewed by a licensed clinician before going live, these are clinical communications, not marketing copy.

Override and feedback

If a staff member reviews the explainability card and **disagrees** (they believe the AI answer IS correct despite the low confidence score) they can mark '**Override: answer is correct' before submitting.** This disagreement event is logged and fed back into the model as a potential false negative. If a staff member's override leads to an approval, the historical rate for that pattern is updated upward.

Feature 4: Staff Feedback Loop, Training the Model in Production (Next Phase)

User story

As the PM, I need every staff interaction with the triage queue to generate a labeled training signal that improves the confidence model over time, so that the model gets better with every PA processed without requiring manual re-labeling by the data team.

Labeled events to capture

Staff action	Signal type	How it improves the model
Bulk approve Green PA → Approved by payer	True positive	Reinforces confidence threshold for this drug/payer/plan. Slight upward adjustment to Green threshold.
Bulk approve Green PA → Denied by payer	False positive	Critical signal. Lowers historical approval rate for this combination. Triggers review of why this PA was Green-tiered.
Staff overrides explainability card → Approved	False negative	The model was too conservative. Updates historical rate upward for this pattern.
Staff edits AI answer → PA approved	Quality improvement signal	Captures what change the staff made. If the same edit pattern recurs, the AI answer generation model is updated.
Staff adds missing doc → PA approved	Documentation gap signal	The specific doc field that was missing becomes a higher-weight completeness requirement for this payer/drug combination.

PM-facing signal dashboard

The PM should have a weekly review dashboard that surfaces: (a) false positive rate (Green-tier PAs that were denied), (b) false negative rate (Red-tier PAs that were approved without edits), (c) top 5 explainability card templates being shown most frequently, and (d) correlation between confidence score tier and actual approval rate. This dashboard is the primary tool for calibrating the model, the PM reviews it weekly in the first 90 days and monthly thereafter.

Feature 5: In-Flight Watch List, Proactive Denial Prevention (Next Phase)

User story

As a PA coordinator, **I need to see which submitted PAs are at high risk of denial BEFORE the payer sends a determination**, so I can proactively collect additional documentation or alert the prescriber while there is still time to influence the outcome.

How it works

The Watch List is a secondary tab in the Triage Queue showing **all PAs that have been submitted and are awaiting determination,** sorted by their confidence score at time of submission. PAs submitted with Red-tier confidence scores appear at the top. Each row shows:

- Confidence score at submission time.
- Days since submission and expected determination timeline.
- Predicted denial reason based on the flagged sub-scores (from the explainability card shown at review time).
- Proactive action: 'Collect BMI documentation', 'Confirm prior therapy dates with prescriber', 'Verify step therapy criteria with payer.'

Escalation workflow

If a Red-tier PA has been in flight f**or 5+ days without a determination**, the system surfaces an escalation prompt: 'No response received in 5 days. Based on this PA's confidence score (28%), consider proactively calling the payer to confirm receipt and clarify denial risk before the formal determination is issued.' This converts the follow-up call from a status check into a clinical intervention call.

2. Confidence Score Algorithm

This section specifies how each of the **four confidence factors i**s computed.

2.1 Factor 1: Historical Approval Rate (Weight: 35%)

Definition

The percentage of prior PA submissions with the same drug (at NDC or brand level), payer, and plan combination that resulted in first-pass approval, over a rolling 12-month lookback window.

Computation

Query: SELECT COUNT(approved) / COUNT(*) AS approval_rate FROM pa_outcomes WHERE drug_ndc = [ndc] AND payer_id = [payer] AND plan_id = [plan] AND submitted_at > NOW() - INTERVAL 12 MONTHS

Data requirements

- Minimum 10 submitted PAs for a given combination to use the historical rate. Below 10: use payer-level rate (same payer, any plan, same drug). Below 10 at payer level: use drug-level rate (any payer, same drug class). Below 10 at drug-class level: default to 0.50 (neutral, 50%) and flag 'Insufficient history.'
- Weight the 12-month window toward more recent submissions: submissions from the last 90 days receive 2x weight. This handles payer criteria changes, recent denials for the same combination are more predictive than older approvals.

Edge cases

- Newly approved drug (FDA approval within 12 months): historical rate defaults to drug-class rate + a 'New drug' flag that lowers overall confidence by 5 points.
- Payer newly integrated (Develop Health's first month with a payer): default to 50% + 'New payer' flag.
- Generic substitution: if brand drug is being submitted but generic equivalent exists in the payer's formulary, surface 'Generic available on formulary, payer may deny brand PA' as an additional flag.

2.2 Factor 2: Documentation Completeness (Weight: 30%)

Definition

The percentage of the payer's known required documentation elements that are present, parseable, and within the acceptable timeframe in the patient's EHR record at the time of PA generation.

Payer criteria library

To compute this sub-score, Develop Health must maintain a structured library of payer PA criteria, specifically: (a) what data elements the payer requires, (b) the format required (ICD-10 code specificity, date range for lab values, dosage/duration for prior therapy), and (c) the acceptable timeframe for each element. This library is the critical data asset that enables the documentation completeness score. It must be maintained by a clinical ops team member and refreshed whenever payer criteria changes are detected.

Computation

For each required element in the payer criteria library for this drug/payer combination: check EHR data at patient level → present and within timeframe = 1.0 / present but expired = 0.5 / missing = 0.0. Documentation Completeness = mean of element scores × 100.

Key required elements by drug class (examples)

Drug class	Typically required elements	Common timeframe requirements
GLP-1 / Obesity	BMI, A1C or FBG, prior metformin trial (dose, duration, failure), ICD-10 E11/E66 specificity	A1C within 6 months, BMI within 12 months, metformin trial 90+ days

Buprenorphine / MOUD	OUD diagnosis (ICD-10 F11.20), PDMP check date, prior treatment history, MOUD prescriber waiver (pre-2023)	PDMP within 7 days, OUD diagnosis documented within 12 months
Biologics (Dupixent, Humira)	Diagnosis specificity (atopic derm, RA), prior conventional therapy (methotrexate, TCS), failure documentation	Conventional therapy trial 12+ weeks, failure documented in clinical note
Specialty psych (antipsychotics)	Diagnosis, prior antipsychotic trials (2+ agents), failure/intolerance documentation	Prior trials within 24 months

2.3 Factor 3: AI Answer Quality (Weight: 25%)

Definition

A measure of how well the LLM-generated PA answers hold up to semantic coherence checks, internal consistency checks, and payer-specific logic validation.

Sub-components

Sub-component	What it checks	Weight within this factor
LLM self-confidence	The LLM's own probability distribution over its generated answer tokens. High entropy = lower confidence.	30%
Step therapy logic check	Deterministic check: does the generated answer correctly reflect the step therapy sequence required by this payer? (E.g., if the payer requires metformin before GLP-1, does the answer document metformin failure?)	40%
Internal consistency check	Does the generated answer contain contradictions? (E.g., claims patient failed Drug A, but Drug A is not in patient's medication history)	20%
Answer completeness	Did the LLM leave any required form fields blank or with placeholder text?	10%

Implementation note

The step therapy logic check and internal consistency check should be deterministic rule systems, not LLM evaluations. Using an LLM to evaluate LLM output creates a correlated failure mode , if the LLM is wrong about step therapy, it is likely to also be wrong in its self-evaluation. Rule-based checks are more reliable for payer logic validation.

> *The PM and ML team must decide on the right split between LLM-based and rule-based quality checks. The general principle: use LLM-based evaluation for fluency and coherence (subjective, hard to rule-encode); use rule-based checks for clinical logic and payer compliance (deterministic, documentable, auditable).*

2.4 Factor 4: Payer Criteria Freshness (Weight: 10%)

Definition

How recently were the payer's step therapy criteria verified, and has any signal of a criteria change been detected since the last verification?

Computation

- *If payer criteria was verified within the last 30 days: 100%*

- *If verified 31–60 days ago: 75%*
- *If verified 61–90 days ago: 50%*
- *If verified >90 days ago or criteria change detected: 0% + mandatory flag*

Payer criteria change detection

Develop Health should implement a lightweight payer monitoring system: weekly automated checks against known payer portals and formulary documents (where machine-readable), plus manual monitoring by clinical ops. When a criteria change is detected: immediately set all affected drug/payer combinations' Payer Criteria Freshness score to 0, elevate all queued PAs for that combination to Yellow or Red, and notify the clinical ops team to update the payer criteria library.

3. Model Training & Validation Using Develop Health's Data Asset

3.1 The Training Data That Already Exists

Develop Health's historical PA outcomes database is the core training asset. Every PA that has been submitted through the platform has associated data that can serve as a training example. The key fields required:

Data field	Why it matters for training	Availability
Drug (NDC + brand)	Primary partition for approval rate history	Available, captured at PA generation
Payer + Plan ID	Primary partition, different plans from same payer can have different criteria	Available, captured at benefits verification
Clinical answers submitted	Required for AI answer quality scoring and retrospective analysis of what was said	Available, PA form content stored
EHR data snapshot at PA time	Required for documentation completeness retrospective , what was available when the PA was generated	Partially available, depends on EHR integration depth
Payer determination (Approved / Denied / More info)	Primary label for historical approval rate	Available, determination logged on receipt
Denial reason code	Critical for mapping denial patterns to specific documentation gaps	Available when returned by payer, coverage varies by payer
Staff edits before submission	Captures what changes staff made to AI answers, signal for AI quality improvement	Available in edit history if change tracking is enabled

3.2 Data Audit, Before Building the Model

Before training, a data audit with the data engineering team to assess:

- **Volume:** How many PA submissions exist per drug/payer/plan combination? Any combination with fewer than 50 submissions should be identified as **'thin data'** and will use fallback rates during initial deployment.
- **Label quality:** What percentage of submitted PAs have a recorded determination? Unresolved PAs (still awaiting determination after 30 days) should be excluded from training.
- **Denial reason coverage:** What percentage of denied PAs have a structured denial reason code vs. a free-text reason? Free-text reasons require NLP parsing before they can be used as training labels.

- **EHR snapshot coverage:** For what percentage of historical PAs is there a complete EHR data snapshot at time of PA generation? This determines whether the documentation completeness factor can be trained retrospectively or must be trained prospectively (on new PA submissions going forward).

> *If EHR snapshot coverage is below 60%, the PM should treat documentation completeness as a prospective-training-only factor for the first model version. Deploy the model with Historical Approval Rate (35%), EHR Snapshot where available (30%), and AI Answer Quality (25%) as the primary factors, and collect EHR snapshots going forward to improve completeness scoring.*

3.3 Training Approach, Phase 1 (Baseline Model)

Historical approval rate component

This is the simplest and most reliable component to train. Compute approval rate for every drug/payer/plan combination with sufficient volume using the historical outcomes database. Store as a lookup table, updated weekly with new outcomes. No ML required, pure statistics.

Documentation completeness component

Build the payer criteria library first. Then retrospectively score historical PAs against the criteria library to determine whether the documentation was complete at submission time. **The approval rate at different completeness levels (80%+ vs. 60–80% vs. below 60%)** validates that completeness is actually predictive , this is the core validation step before assigning its weight.

AI answer quality component

Use the LLM's internal log probabilities for the self-confidence sub-score. Build deterministic rule systems for step therapy logic check and internal consistency check. Validate by comparing these quality scores against the actual approval outcome, high-quality score should correlate with higher approval rate.

3.4 Validation Protocol

Before releasing the confidence score to production, run an offline back-test using the historical outcomes database:

Validation test	Method	Pass criteria
Green-tier calibration	Of all historical PAs the model would have scored Green (75+), what % were actually approved?	Target: 88%+ actual approval rate for Green-tier PAs. If below 88%, raise the Green threshold or reduce Green weights.
Red-tier precision	Of all historical PAs the model would have scored Red (<40), what % were actually denied?	Target: 70%+ actual denial rate for Red-tier PAs. Below 70% = too many false alarms, eroding staff trust.
Subgroup fairness	Does model performance degrade for specific patient demographics, drug types, or payer categories?	No subgroup should have **>10 percentage point difference in false positive rate vs. overall.**
Stability test	Does the model score the same PA consistently when re-run with the same inputs?	100% determinism for Historical and Documentation factors. <5% variance for AI quality factor.

Human validation step

Before the model goes live, have 2–3 experienced PA coordinators review 50 randomly sampled PAs and estimate the confidence they would assign based on their own clinical experience. Compare their estimates to the model's scores. **Correlation should be above 0.65.** Where they disagree significantly, investigate , the clinical staff's intuition captures signals the model may miss.

3.5 Ongoing Model Improvement, Production Learning Loop

The model improves through four channels in production:

1. **Weekly outcomes ingestion:** every PA determination received from payers is ingested into the outcomes DB and updates the historical approval rate lookup table.
2. **Staff feedback signals:** bulk approve events, override events, edit events. Processed weekly.
3. **Payer criteria library updates:** when clinical ops updates the criteria library after a payer change, all affected PAs in the queue are immediately re-scored.
4. **Quarterly model recalibration:** review the false positive and false negative rates (from the PM dashboard), adjust factor weights, and re-run validation tests. Schedule for the end of each quarter.

4. Prototype

The goal of the prototype is to **demonstrate the core value proposition, that confidence scoring dramatically changes how staff process the PA queue in a form**

4.1 Success Metrics

These should be measured in the 60–90 days after production launch:

Metric	Target	Why it matters
Green-tier PA review time	<90 seconds per PA (from 4.5 min baseline)	Directly measures staff efficiency gain. Measurable via session analytics on the review UI.
False positive rate (Green PAs denied)	<12% of Green-tier approvals result in payer denial	Measures trust calibration. Above 12% erodes staff trust in the Green tier.
Overall approval rate (network)	>2 percentage point improvement vs. pre-launch baseline	Measures whether the intervention on Red/Yellow PAs is improving clinical documentation quality , the upstream flywheel.

The PA Confidence Score is not a feature, it is the product interface of Develop Health's core data moat. The UI described here is the first surface where that moat becomes tangible to customers. It allows every PA that flows through the system from day one becomes a data point that makes the next version of the model better. No competitor can replicate this without going through the years of PA volume that Develop Health has already accumulated.

How to read Mermaid Chart

Zone 1) PA Initiation: The trigger. A prescription is written, Develop Health detects PA is required, pulls the EHR snapshot, and the AI generates the form and clinical answers. Feeds into Zone 2.

Zone 2) Confidence Score Computation: The core. Four signals (historical approval rate, documentation completeness, AI answer quality, payer criteria freshness) combine into a 0–100 score. Score maps to a tier. Explainability card generated for Yellow and Red. Feeds into Zone 3. Receives improvement signals from Zone 6.

Payer Criteria Library: A standing data store owned by clinical ops. Supplies required documentation elements and step therapy rules to Zone 2, and receives formulary change alerts from Zone 5.

Zone 3) Triage Queue: Staff action layer. Green PAs bulk- approved in under 30 seconds. Yellow PAs reviewed with the explainability card. Red PAs blocked→ staff collects missing data and requeues. All approved PAs feed into Zone 4.

Zone 4) Payer Submission and Determination: Reviewed PAs submitted to the payer. An in-flight watch list monitors Red and Yellow submissions. Payer returns Approved or Denied. Both outcomes feed into Zone 5 as labeled training data.

Zone 5) Feedback Loop: Four input streams→ payer outcomes, staff interaction events, formulary change alerts, and quarterly recalibration → are captured here. Each stream feeds a specific improvement channel in Zone 6.

Zone 6) Continuous Model Improvement: The compounding layer. Approval rates, AI quality weights, payer criteria, and tier thresholds are each updated from Zone 5 signals. The improved model feeds back into Zone 2 making every subsequent PA scored more accurately than the last.